SAFETY PROJECTS PVT LTD
S P P L

We know what protects you the best!

SafePro Fixed Lifeline System

SafePro Roof Lifeline System
SafePro Overhead Lifeline System
Vertical Fall Arrest

www.portcastle.in

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EN 795:2012 Type C | Type A
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SafePro Fixed Lifeline System

A fixed lifeline system is a crucial component of fall protection in various industries, such as construction, roofing, and maintenance. This system is designed to provide a secure anchor point for workers who perform tasks at heights. It typically consists of a permanent horizontal or vertical cable or rail that is securely installed on structures like rooftops, industrial equipment, or buildings.

The fixed lifeline system serves as a lifeline for workers, enabling them to attach their fall protection equipment, such as harnesses and lanyards, to the line. This ensures that in the event of a fall, the worker is safely secured, preventing them from plummeting to the ground.

Fixed lifeline systems are a proactive safety measure, designed to mitigate the risks associated with working at heights. They offer peace of mind to both employees and employers by reducing the potential for accidents and injuries, ensuring compliance with safety regulations, and promoting a safer work environment.

SafePro Fixed Lifeline systems are tested and certified as per work at height regulations.

Application Standards

EN 795:2012
This European standard specifies the requirements for performance and associated test method for single user anchor devices. The standard distinguishes 5 types of anchor devices.

Type A:
Single anchorage points which are firmly anchored to the substructure

Type B:
Temporary anchorage points that can be removed again after usage without leaving any traces

Type C:
Flexible anchor which deviates horizontally by not more than 15° (Lifeline Systems)

Type D:
A rigid anchor which deviates horizontally by not more than 15° (Rigid Rail safety systems)

Type E:
Anchorage devices that are held in place by their own weight

CEN/TS 16415:2013
This technical specification certifies anchor devices that allow more than one user to be attached at the same time. Since work at height requires at least two workers, SafePro tests all systems for several people.

EN 353-1:2018
This European standard defines the technical requirements for vertical rigid anchor line (rail or wire rope under tension). The forwardleaning angle and/or sideways leaning angle lies between 85° and 95°. 2006/EC/42:2006

- **Industrial Units**
- **Buildings**
- **Over Trucks**
- **Vertical Structures**
- **Fall Arrest**
- **Rope Access**
- **Work in restraint**
- **Confined Spaces**
- **Wall configuration**
- **Potentially explosive atmosphere**

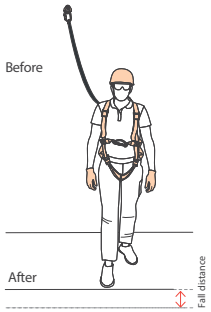
Fall Calculation

Fall Factor

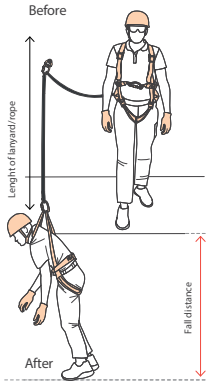
The fall factor is the distance fallen, divided by the length of the lanyard or rope available to absorb that fall. The fall factor ranges from 0 to 2. Independently on a fall being long or short, the lower the fall factor, the safer the fall.

$$\text{Fall Factor} = \frac{\text{Fall distance}}{\text{Lenght of lanyard/Rope}}$$

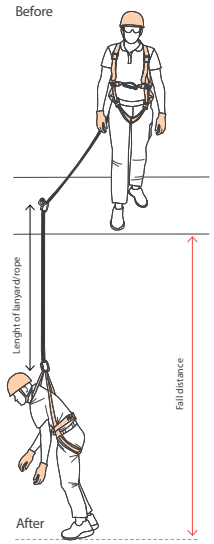
→ Fall Factor 0



→ Fall Factor 1



→ Fall Factor 2



Fall Clearance

The height of fall must be less than the available clearance so that the person falling does not hit an obstacle during the fall. The height of fall is the sum of the terms given below:

D Deflection: Deflection of the rigid anchor line.

Ll Lanyard: Length of the lanyard (<2m).

La Absorber: the braking distance of the energy absorber.

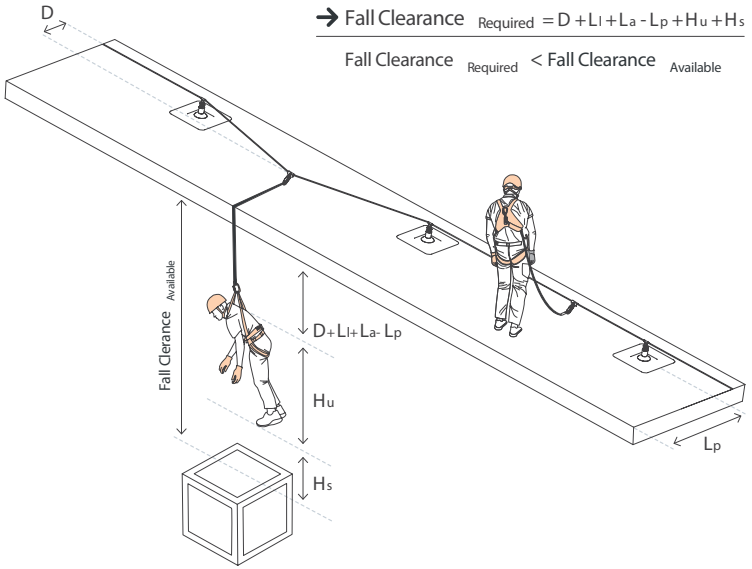
Lp Path: Distance between the lifeline and the edge from which the user can fall

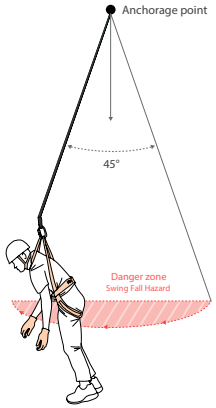
Hu: User height

Hs: Safety distance = 1m

→ Fall Clearance $_{\text{Required}} = D + Ll + La - Lp + Hu + Hs$

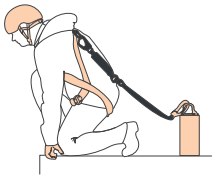
Fall Clearance $_{\text{Required}}$ < Fall Clearance $_{\text{Available}}$



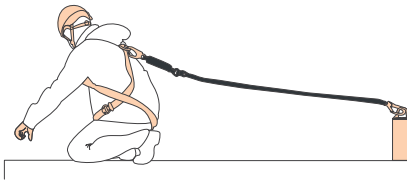


Some types of work at heights may include a risk of pendulum effect, causing the fallen worker to swing into obstacles. A maximum angle of 30° with respect to the anchorage system or anchor point is recommended to minimize that pendulum effect.

→ Fall Arrest



→ Fall Restraint



Fall restraint system prevents a person from reaching the fall risk and a fall arrest system minimises injury in the event of a fall.

Where Innovation meets safety

Ideation
It is the creative spark that ignites innovation and lays the groundwork for further development.

Strategy & Planning
Evaluate and refine the generated ideas, establish goals, allocate resources, and create a strategic roadmap to metamorphose those ideas into actionable plans.

Prototyping
To visualize the design in a realistic manner, identify potential issues, and refine the product.

Execution
Our Execution is the practical realization of plans and strategies, turning concepts into tangible results.

Quality Check
Quality check, a meticulous blend of scrutiny and testing, ensures that products or services adhere to impeccable standards and specifications, leaving no room for imperfections to mar their debut before discerning customers or users.

SafePro Service Solutions

Installation Services

Inspection and Testing Services

Maintenance and Repair Services

Training and Education

Consultation and Risk Assessment

Fall Protection System Design

Documentation and Record-Keeping

Quality Assurance

Quality is the cornerstone of our commitment. Every safety solution we provide is meticulously tested and scrutinized to ensure it meets the highest industry benchmarks. We take pride in delivering safety solutions and continuous after sale services that consistently exceed industry standards.

SafePro
Roof Lifeline System

Metal Roof Sheet Mounted

Compliant with

CEN/TS 16415:2013 C

EN 795: 2012 C

SafePro Roof Lifeline System is a critical safety apparatus designed for workers who need to access and perform tasks on roofs, elevated structures, or other elevated workspaces to prevent falls and minimize the risk of injury or fatality. This system consists of a series of anchor points, cables, and connectors installed horizontally along the rooftop allowing workers the freedom to move horizontally across the roof while remaining securely attached to the lifeline at all times.

Application Areas

- Shuttering on construction sites
- Building Inspection
- Solar Panel Installation and Maintenance
- Roof repairs and maintenance
- Roofing and structural work on elevated surfaces

Potential Hazards

Features

Fully traversable and continuous fall protection system: Pass freely over all brackets & corners, allowing access and egress anywhere along the system and avoiding the need for expensive entry/ exit terminals.

Multi-user: Up to 4 users across a maximum span of 15 m depending on the line configuration.

Multi-purpose: Fall arrest, work restraint, and work positioning.

Anti – corrosion: Made of corrosion resistant stainless steel which is Compatible with adverse working conditions, ranging from -40 degrees centigrade to +110 degrees centigrade.

Dual shock absorbing system: "DecoTECH energy dissipating technology" and additional shock absorber in line

.

Versatile applications: Modular post and universal base plate - covers virtually all standard application and are tested in conjunction with major roofing manufacturers of trapezoidal, standing-seam, and Kalzip.

Centralized system: Posts may be installed as single point anchors for centralized maintenance tasks within a specific area of interest

DecoTECH™
Technology
SAFEPRO
with DecoTECH

Distinctive Innovation

DecoTECH Energy Dissipating Technology

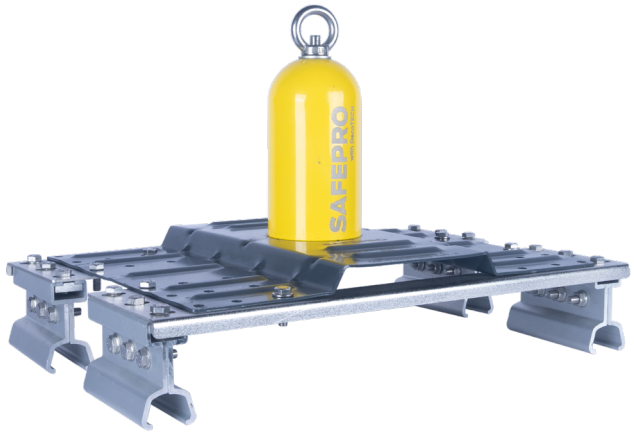
Applied for patent

SafePro roof anchors are designed to enhance fall protection systems on roofs by dissipating and reducing the forces generated during a fall. These anchors are crucial for maintaining the structural integrity of the roof and protecting individuals working at heights.

These posts typically consist of a sturdy structure that can securely anchor a worker's fall protection system and they incorporate mechanisms that absorb and dissipate the forces.

Fall Protection from Any Angle

Regardless of the direction or angle from which a fall happens, the 360-degree energy absorbing deploying feature ensures that the forces are mitigated, protecting the user and the roof structure. This is crucial because falls can occur in various orientations, and the anchor needs to be effective regardless of the angle.



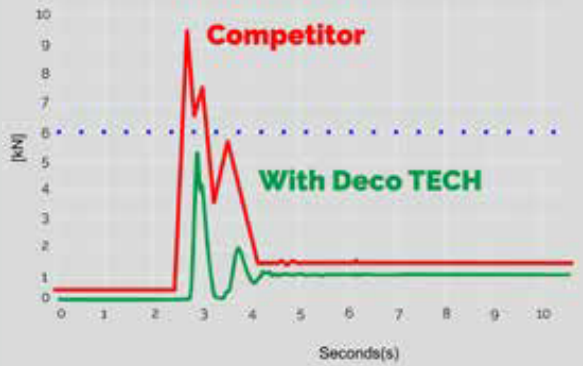
Ensuring integrity of roof with DecoTECH

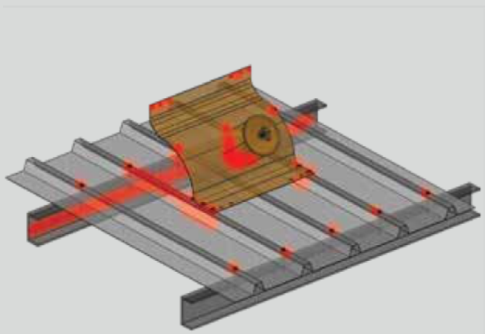
Applied for patent

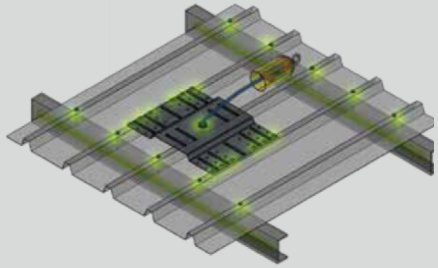
In case of a fall, the SafePro anchor activates its innovative DecoTECH Energy Dissipating Technology. This technology effectively absorbs and dissipates forces, ensuring that the impact on the roof structure is minimal, reducing it to less than 6 kN. This force reduction is the most impressive in the market, making the anchor suitable for installation on various roof types, whether they are old or new. This versatility also enhances safety by minimizing the risk to the structural integrity of the roof.

When dynamic drop test is conducted these forces get generated

DecoTECH vs Competitor



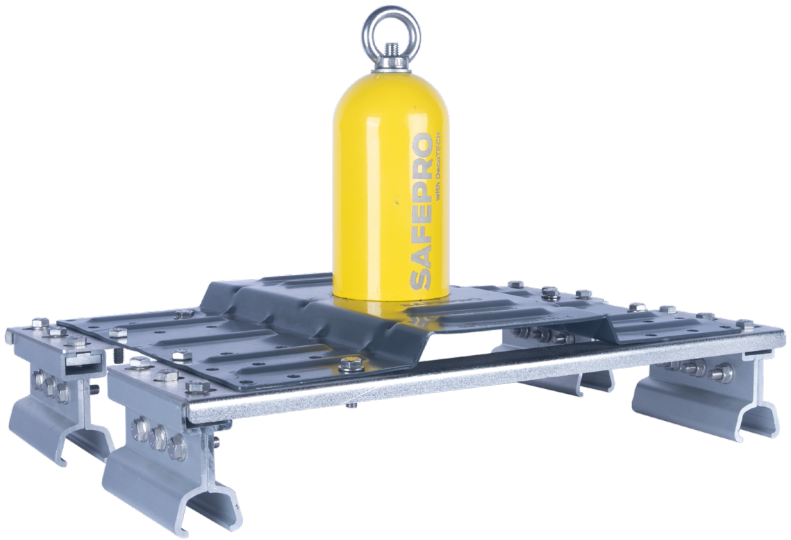




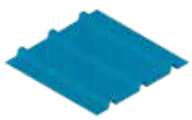
In event of a Fall

Stress Concentration

- High stresses on substructure
- Minimised stresses on substructure due to DecoTECH technology

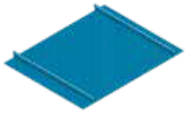


Additional fixing accessories to fit on diffrent sub-structures.



Corrugated Roof Sheet

Standing Seam Roof Sheet

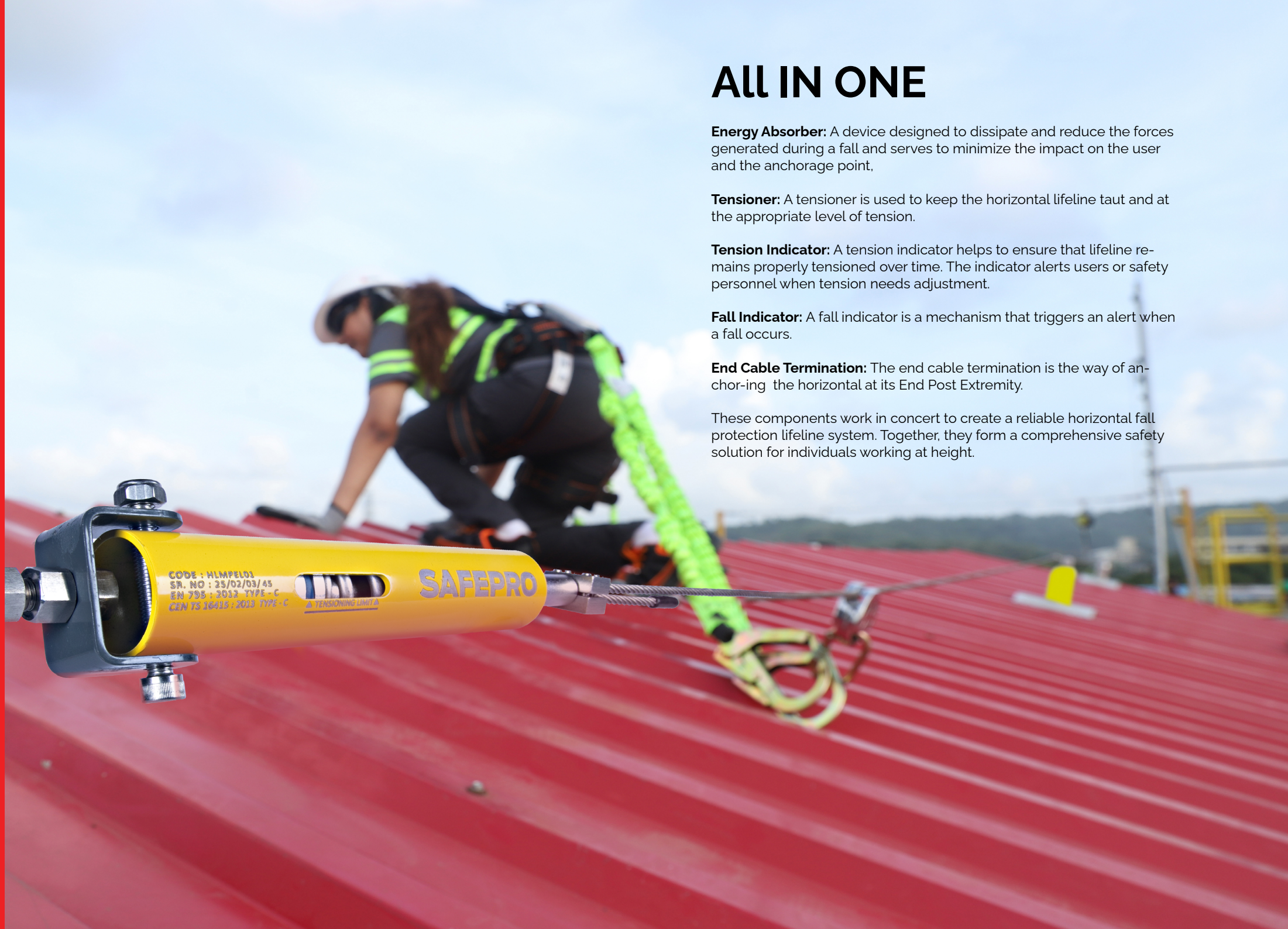


Kalzip Roof Sheet

Features

- Ease of installation
- Adjustable
- Versatility
- Corrosion Resistant
- Load Distribution

“Work Safe, Stay Safe”
SAFEPRO



All IN ONE

Energy Absorber: A device designed to dissipate and reduce the forces generated during a fall and serves to minimize the impact on the user and the anchorage point,

Tensioner: A tensioner is used to keep the horizontal lifeline taut and at the appropriate level of tension.

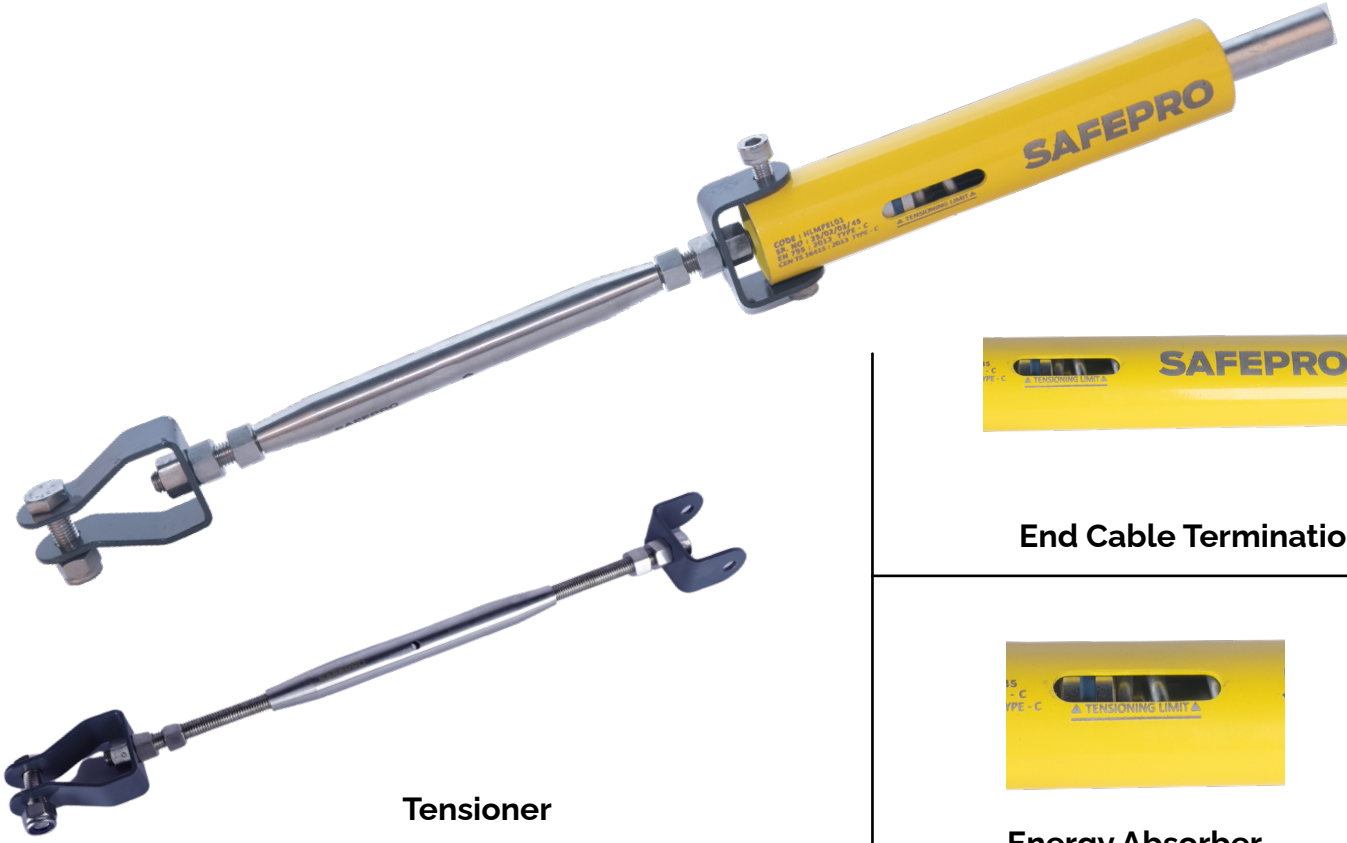
Tension Indicator: A tension indicator helps to ensure that lifeline remains properly tensioned over time. The indicator alerts users or safety personnel when tension needs adjustment.

Fall Indicator: A fall indicator is a mechanism that triggers an alert when a fall occurs.

End Cable Termination: The end cable termination is the way of anchor-ing the horizontal at its End Post Extremity.

These components work in concert to create a reliable horizontal fall protection lifeline system. Together, they form a comprehensive safety solution for individuals working at height.

Essential Features of All in One



Tensioner

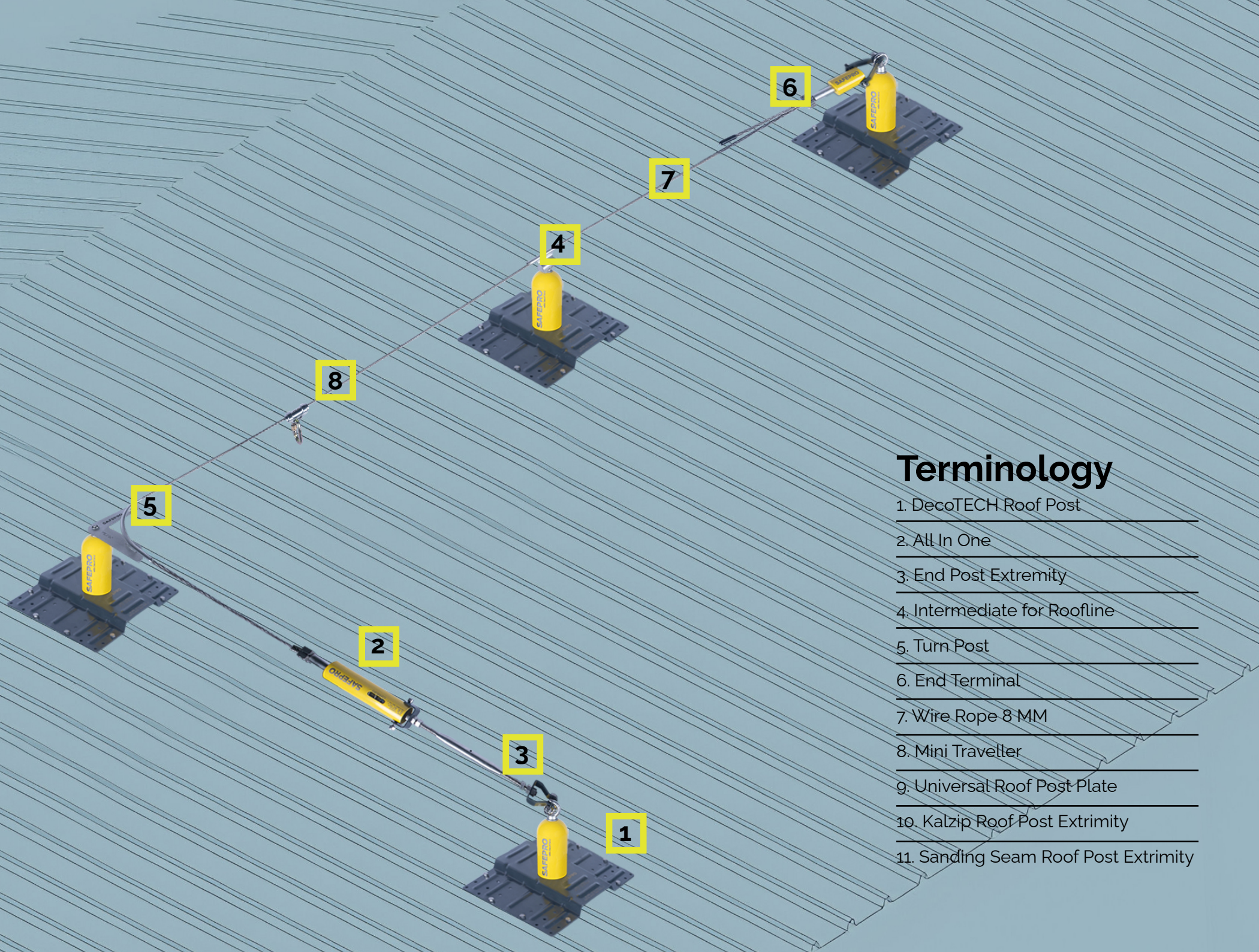
End Cable Termination

Energy Absorber

Tension Indicator

Red marking
indicate a fall

Fall Indicator



Terminology

1. DecoTECH Roof Post
2. All In One
3. End Post Extremity
4. Intermediate for Roofline
5. Turn Post
6. End Terminal
7. Wire Rope 8 MM
8. Mini Traveller
9. Universal Roof Post Plate
10. Kalzip Roof Post Extrimity
11. Sanding Seam Roof Post Extrimity

Roof Lifeline Components

DecoTECH Roof Post

All In One

Intermediate for Roofline

Turn Post

End Terminal

Wire Rope

Mini Traveller

Universal Roof Post Plate

Kalzip Roof Post Extremity

Sanding Seam Roof Post Extrimity

End Post Extremity

SafePro
Roof Lifeline System

Asbestos / Cement Roof Sheet Mounted

Compliant with

CEN/TS 16415:2013 C

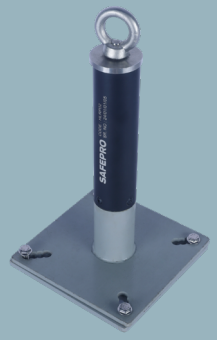
EN 795: 2012 C

SafePro Roof Lifeline System is a critical safety apparatus designed for workers who need to access and perform tasks on roofs, elevated structures, or other elevated workspaces to prevent falls and minimize the risk of injury or fatality. This system consists of a series of anchor points, cables, and connectors installed horizontally along the rooftop allowing workers the freedom to move horizontally across the roof while remaining securely attached to the lifeline at all times.

Application Areas

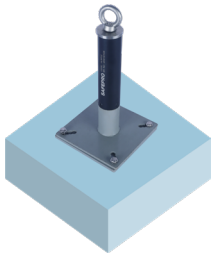
- Roofing and structural work on elevated surfaces
- Asbestos Roof Inspection and Maintenance
- Asbestos Roof Removal
- Asbestos Roof Repair and Sealing

Potential Hazards:



Fully traversable and continuous fall protection system - Pass freely over all brackets & corners, allowing access and egress anywhere along the system and avoiding the need for expensive entry/ exit terminals.

Compatibility with substructures



Concrete Structure

Steel Structure

“Secure steps,
SafePro leads the way”
SAFEPRO

SafePro Overhead Lifeline System

Compliant with

CEN/TS 16415:2013 C

EN 795: 2012 C

SafePro overhead lifeline system is a vital safety tool created to safeguard workers working at heights. By providing a secure anchor point above, it enables individuals to move freely while working on elevated surfaces, transportation, oil and gas, utilities and mining industries gives us a unique advantage in overcoming the challenges. This system is essential for preventing falls and ensuring the safety of personnel engaged in tasks at elevated locations.

Application Areas

- Loading and unloading of vehicles
- Tank filling and loading
- Maintenance of aircraft
- Maintenance & repair of transportation vehicles, trains and carriages
- Maintenance of manufacturing and distribution equipment such as conveyors
- Overhead crane access

Potential Hazards:

Featurest

Flexibility: Wire rope based systems offer unparalleled flexibility. Our engineers are able to develop and install both single and multi-span overhead horizontal lifelines that accommodate any railcar or truck application.

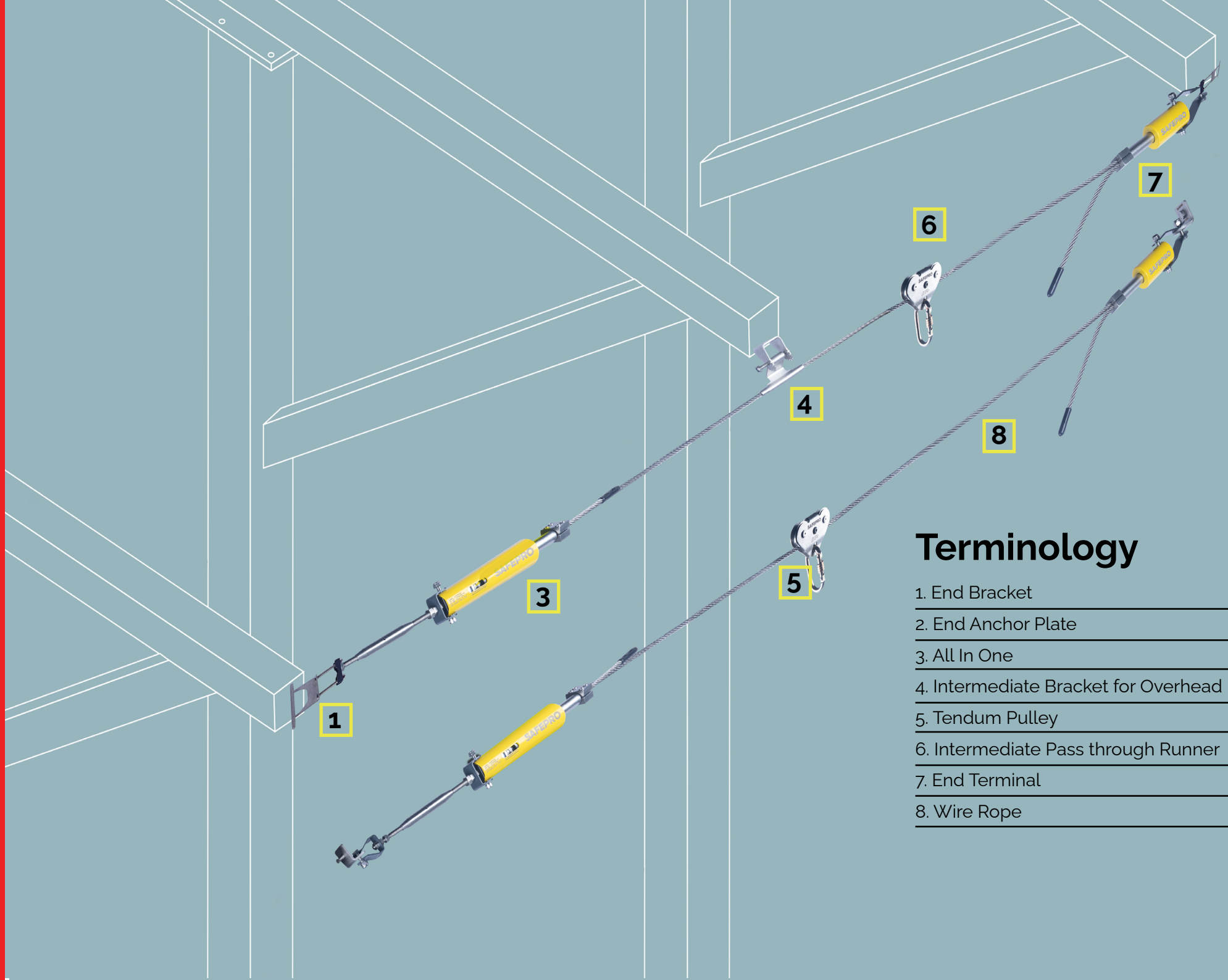
Multi User: Cable based systems allow multiple workers a comfortable and safe access to all working areas of the aircraft. Maintenance personnel will able to safely and efficiently access the entire length of any truck or railcar.

Continuous Fall Protection: The unique and patented overhead shuttle unit glides freely over stainless steel cables providing continuous protection over the entire length of the system.

Engineered Efficiency: Wire rope based systems include built-in user-friendly ALL IN ONE which includes tension indicators, energy absorbers to reduce end loads, serial numbering of all components and are simple and economical to install.

All-Weather Use: Systems are all-weather use for optimum productivity.

“Rise to Safety
Standards”



Terminology

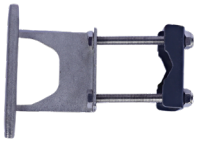
1. End Bracket
2. End Anchor Plate
3. All In One
4. Intermediate Bracket for Overhead line
5. Tendum Pulley
6. Intermediate Pass through Runner
7. End Terminal
8. Wire Rope

Overhead Lifeline Components

Intermediate Bracket
for Overhead line

Intermediate Pass through
Runner

Tandem Pulley



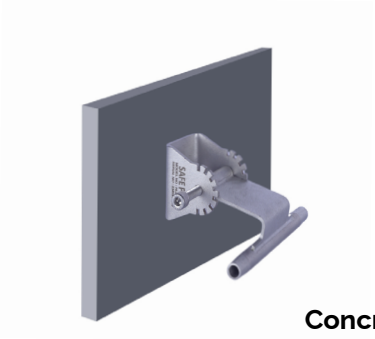
End Bracket

End Anchor Plate

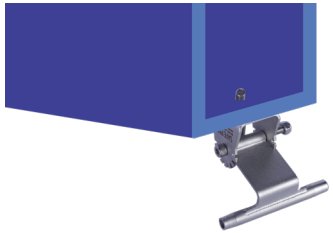
Wire Rope

End Terminal

Compatibility with sub-structures

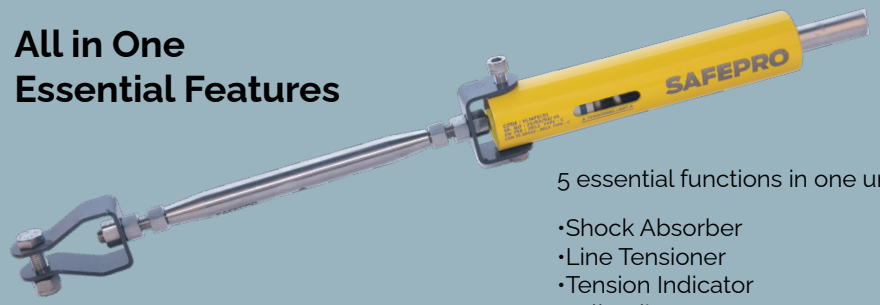


Concrete Structure



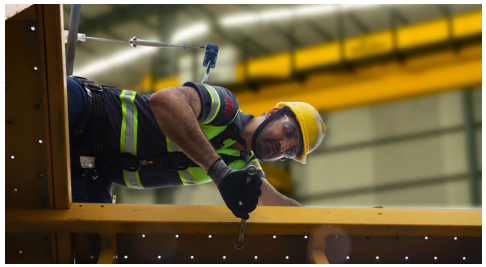
Steel Structure

All in One Essential Features



5 essential functions in one unit

- Shock Absorber
- Line Tensioner
- Tension Indicator
- Fall Indicator
- Cable End Termination



Crane bay

Pipe Rack

SafePro Vertical Lifeline System

Compliant with

EN 353-1

SafePro Vertical Lifeline System (VLS) is a safety device that can be retrofitted onto existing climbing devices such as ladders or rungs as a fall protection measure in various industries, ensuring worker safety and preventing accidents.
It typically consists of a vertical cable or rope securely anchored at the top, allowing individuals to attach themselves and move safely while performing tasks at elevated positions.

Potential Hazards:

Application Areas:

- Fall arrest
- Maintenance of electrical towers
- Maintenance of wind masts
- Working in confined spaces
- Maintenance of industrial, residential & airport roofs
- Maintenance of commercial office and mall roofs

Features

Corrosion-resistant: Stainless steel construction.

Quick and easy to install: Absolute single-hand operation.

Integrated mechanism prevents upside-down installation of the cable sleeve to the cable.

Hands-free trailing: While climbing and descending, and automatically bypasses intermediate Automatic Pass-Through Cable Guides.

Spring activated mechanisms: Always move device back to initial position.

Dual locking mechanism: Prevents accidental un-locking of the device from the cable.

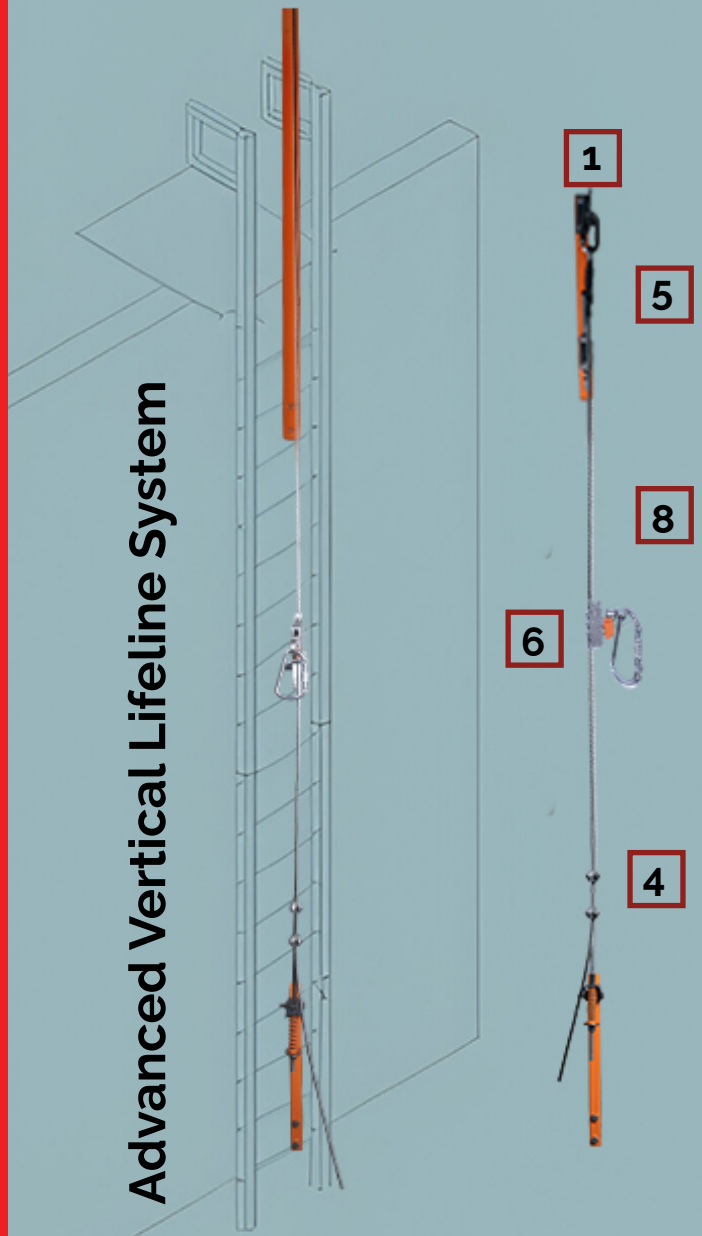
Instantly locks

Built-in metal shock absorber: Reduces impact forces on the user in the event of a fall.

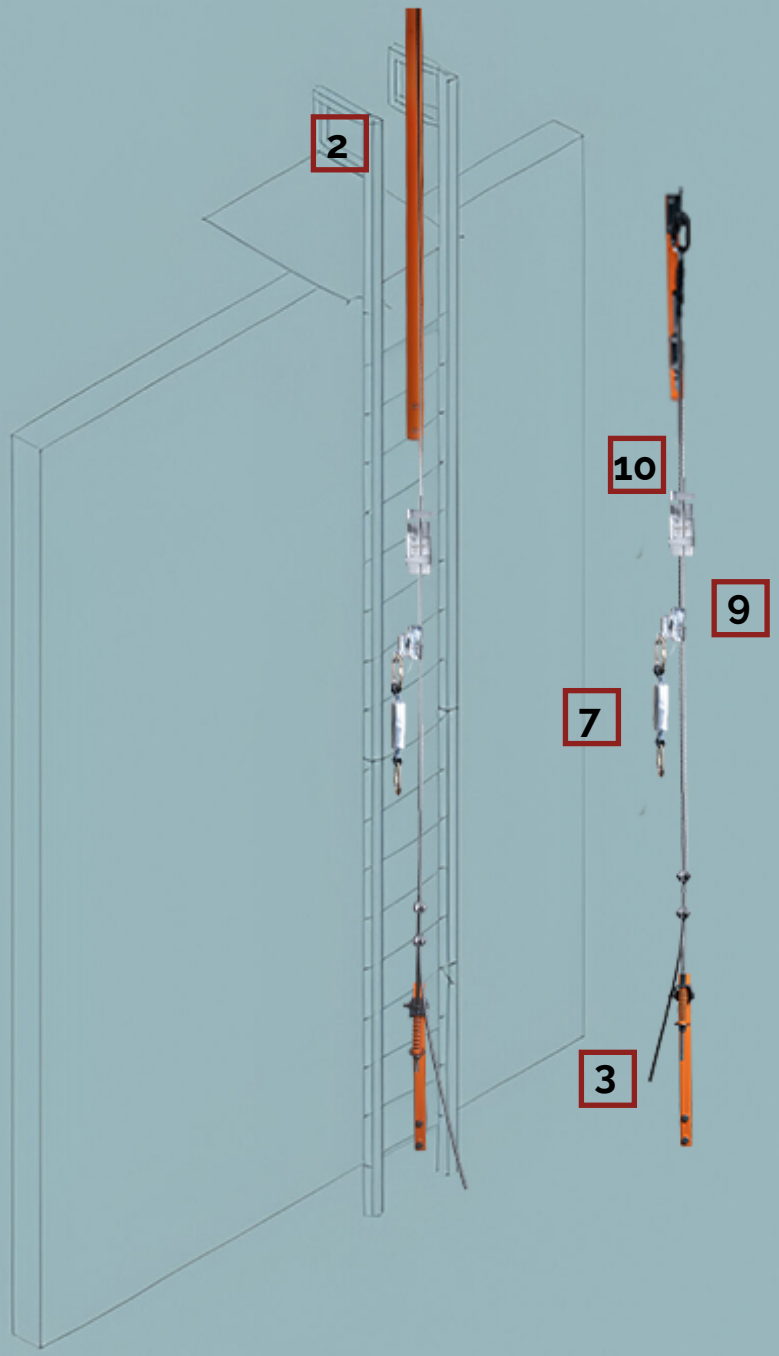
Fall indicator provides for ease of inspection.

“Climbing safely,
thriving confidently.”

Advanced Vertical Lifeline System



Basic Vertical Lifeline System



Terminology

1. Top Bracket
2. Extended Bar
3. Bottom Bracket
4. Tensioning Stud
5. Cable Extremity
6. Advanced Intermediate Bracket for vertical line
7. Basic Intermediate Bracket
8. Automatic Pass Through Fall Arrester
9. Fall Arrester with webbing energy absorber
10. Wire Rope

Fall Arrester Devices

Fall arrester includes a mechanism that locks onto a lifeline or anchor point in the event of a sudden drop, providing protection against serious injuries or fatalities while working at heights.

Automatic Pass through Fall Arrester

- Integrated Energy Absorber
- Anti inversion mechanism
- Intermediate Pass-through
- No Maneuvering
- One hand Operation

Fall Arrester with Webbing energy absorber

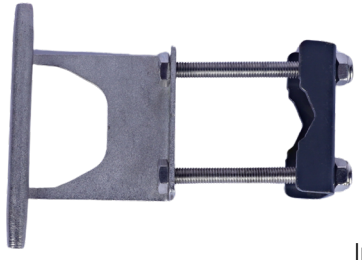
Vertical Lifeline Components

Top Bracket

Bottom Bracket

Tensioning Stud

Cable Extrimity



Advanced
Intermediate Bracket

Wire Rope

Basic
Intermediate Bracket

Compatibility on Substructure

Telecom Totwers.

Transmission Towers

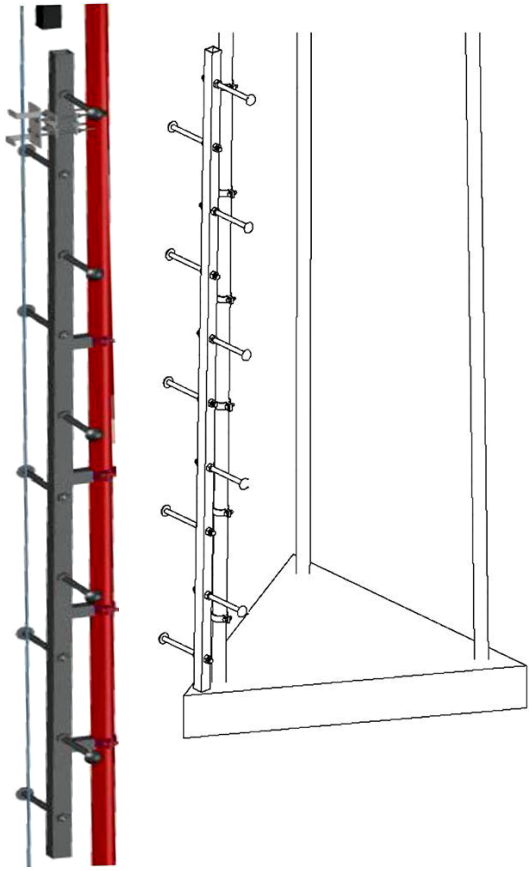
Industrial Ladders

Chimney Ladders

Engineered Fall Protection System

Vertical
Access System

- The tower structure will be equipped with Ladder on the external leg with fall protection system on it
- The Access system can be mounted on any type of sub structure like, Round Sections, Square Hollow Sections, Rectangular Hollow Sections, Angles and Channels



Potential Hazards:

Main System Component

Masts

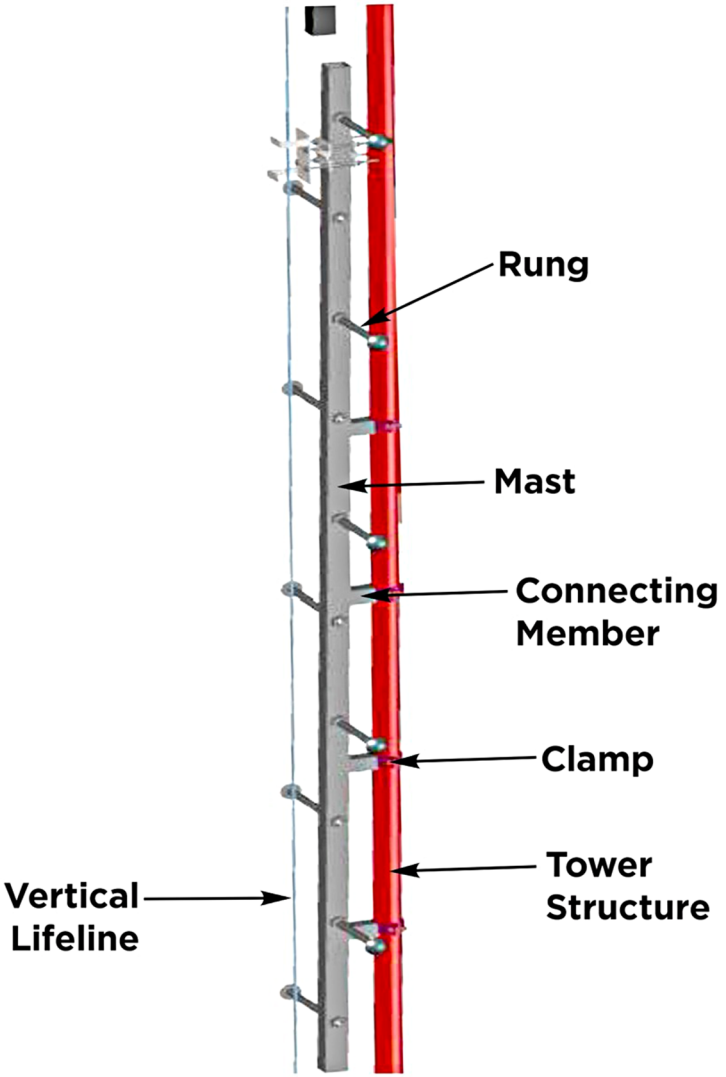
Clamps

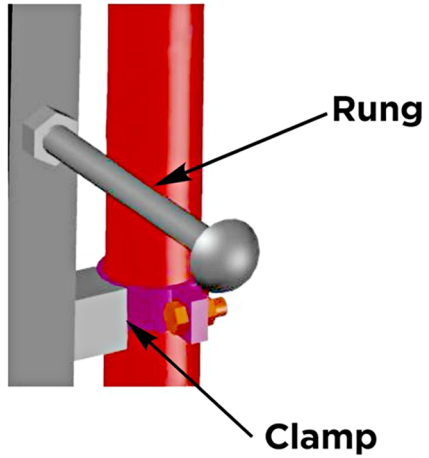
Connecting Members

Rungs

Vertical Lifeline System.

Nomenclature





Fall Protection on Vertical Access System.

- Top Bracket is installed on the top segment of the access system.
- Intermediate Bracket is installed at every 10 m span.
- Bottom Bracket is installed on the bottom segment of access system.
- This system will come with Fall Arrester with Webbing Energy Absorber

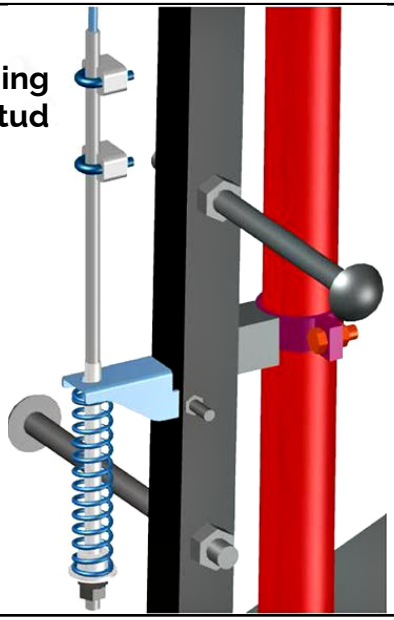
Top Bracket

Fall Arrester with Webbing Energy Absorber

OR

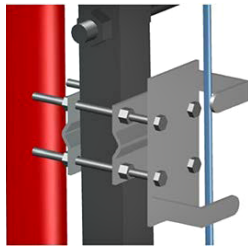
Automatic Pass through Fall Arrester

- Integrated Energy Absorber
- Anti inversion mechanism
- Intermediate Pass-through
- No Maneuvering
- One hand Operation



Tensioning Stud

Bottom Bracket



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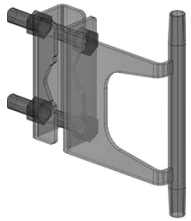

Intermediate Bracket

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